Minor Project
"Automatic Number Plate Recognition System"

A Project report submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING (CSE)

With

Artificial Intelligence and Machine Learning

Submitted to

SANDIP UNIVERSITY, SIJOUL

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This is to certify that the project report entitled **"Automatic Number Plate Recognition System"**, is a bonafide record of the work done by **"Raju Kumar"** bearing the **"PRN:240205241033"** , in the partial fulfilment of the requirements for the award of Batchlor of Technology in Computer Science and Engineering in the School of Computer Science and Engineering, Sandip University, during the year "**2025-26**". The work content in this report has not been submitted in any other University/Institute for the award of any degree/diploma.

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I hereby declare that the project report entitled **"Automatic Number plate Recognition System",** submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (CSE)** is my original work carried out under the guidance of **Dr. Tauseef Ahmad**

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The project report entitled **"Automatic Number Plate Recognition System"**, submitted by **"Raju Kumar"** bearing **PRN No:2402025241033**, has been examined by us and is hereby approved for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (CSE)** at **Sandip University**.
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(Internal Examiner) **(External Examiner)**

Date: **Date:**

ACKNOWLEDGEMENT

I express my sincere gratitude to **Dr. Tauseef Ahemad**, for his valuable guidance, constant encouragement, and support throughout the development of this project. Without his continuous assistance and supervision, the successful completion of this work would not have been possible.

I would also like to express my sincere thanks to **Prof. Sonal Kumar**, Head of the Department of Computer Science and Engineering, for providing the necessary facilities and support to carry out this project work.

I am grateful to all the faculty members and staff of the Department of Computer Science and Engineering for their direct and indirect assistance during the course of this project.

Finally, I would like to thank my friends and my parents for their continuous encouragement, motivation, and support during my academic journey at **Sandip University, Sijoul**.

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Abstract

Automatic Number Plate Recognition (ANPR) is an important application of computer vision that enables automatic detection and identification of vehicle license plates from images. With the increasing number of vehicles, manual monitoring systems have become inefficient, creating a need for automated and accurate solutions.

This project presents an ANPR system developed using **Python, OpenCV, and EasyOCR**, designed to detect and recognize Indian vehicle number plates. The system first applies image preprocessing techniques such as grayscale conversion and noise reduction, followed by edge detection and contour analysis to locate the number plate region. After isolating the plate, Optical Character Recognition (OCR) is used to extract the textual information from the image.

Unlike traditional approaches that rely on template matching, this system uses modern OCR techniques to improve flexibility and adaptability across different plate formats. Although the system may not achieve 100% accuracy due to variations in lighting, angle, and image quality, it provides reliable and efficient results for real-world applications such as traffic monitoring, parking systems, and security surveillance.

Keywords: OpenCV, EasyOCR, Edge Detection, Contour Detection, Image Processing, ANPR System

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Chapter 1
Introduction

1.1 Background

Automatic Number Plate Recognition (ANPR) is a key application of computer vision used for identifying vehicle registration numbers automatically. With the rapid growth in the number of vehicles, traditional monitoring systems are becoming inefficient.

Currently, in many places:

- Vehicle number identification is done manually
- Requires human effort and continuous monitoring
- Prone to errors and delays
- Not suitable for large-scale traffic systems

ANPR systems help automate this process by using image processing and Optical Character Recognition (OCR) techniques to detect and read number plates from images or videos.

1.2 Problem Statement

Manual vehicle number plate identification is time-consuming, inefficient, and error-prone. In areas such as traffic control, parking management, and security systems, there is a need for an automated solution that can quickly and accurately detect and recognize number plates from images.

1.3 Objectives

The main objectives of this project are:

- To develop an automated number plate detection system
- To extract number plate region using image processing techniques
- To recognize characters using OCR (EasyOCR)
- To display detected number along with the image
- To improve efficiency and reduce manual effort

1.4 Scope of the Project

The scope of this project includes:

- Processing static vehicle images
- Detecting number plate using contour-based methods

- Extracting and recognizing text using OCR
- Displaying results with bounding box and detected text

1.4.1 Limitations:

- Works best with clear images
- Accuracy may reduce with blur, angle, or low lighting
- Not optimized for real-time video (basic version)

1.5 Organization of Report

This report is organized as follows:

- **Chapter 1:** Introduction to the project
- **Chapter 2:** Literature Review and related work
- **Chapter 3:** System Design and Methodology
- **Chapter 4:** Implementation and Results
- **Chapter 5:** Conclusion and Future Scope
- **Chapter 6 :** System Testing
- **Chapter 7:** Results and Discussion
- **Chapter 8:** Conclusion and Future Work
- **References**
- **Appendix**

Chapter 2
Existing System and Literature Review

2.1 Existing System

Automatic Number Plate Recognition (ANPR) systems have been widely used in developed regions for applications such as traffic monitoring, toll collection, and security surveillance. However, in many areas, especially in developing regions, vehicle identification is still largely dependent on manual processes or semi-automated systems.

In the existing system:

- Vehicle number plates are often checked manually by authorities
- CCTV footage is reviewed manually to identify vehicles
- Some automated systems use basic image processing techniques
- Traditional OCR systems like Tesseract are used with limited accuracy

These systems are not fully efficient when dealing with large volumes of vehicles or varying environmental conditions.

2.2 Problems in Existing System

The current systems face several challenges:

- Time-consuming process due to manual verification
- High dependency on human effort
- Prone to errors in reading number plates
- Poor performance in low light or bad weather conditions
- Difficulty handling different plate formats (especially Indian plates)
- Low accuracy in traditional OCR systems

2.3 Limitations of Existing System

- Slow processing speed for large datasets
- Lack of automation in many systems
- Inconsistent accuracy due to varying image quality
- Difficulty in detecting plates from angled or complex backgrounds
- Limited adaptability to different number plate styles
- High cost for advanced commercial ANPR solutions

2.4 Literature Review

Several researchers have proposed different approaches for number plate recognition using computer vision and machine learning techniques.

- Early systems were based on edge detection and morphological operations to locate number plates, followed by traditional OCR engines such as Tesseract. These methods worked well in controlled environments but struggled with real-world variations.
- Later approaches introduced machine learning and deep learning techniques, such as Convolutional Neural Networks (CNNs), for more accurate detection and recognition. These models improved performance but required large datasets and high computational power.
- Some modern systems use YOLO (You Only Look Once) for real-time object detection, which significantly improves plate detection accuracy and speed.
- OCR technologies have also evolved, with tools like EasyOCR providing better accuracy and support for multiple languages without requiring extensive training.

Despite these advancements, achieving perfect accuracy remains a challenge due to variations in lighting, angle, resolution, and plate formats.

2.5 Proposed Solution Overview

To overcome the limitations of existing systems, this project proposes an ANPR system using OpenCV and EasyOCR.

The developed system:

- Uses image preprocessing techniques to enhance image quality
- Applies edge detection and contour analysis to locate the number plate
- Extracts the plate region using masking techniques

- Uses EasyOCR for accurate text recognition
- Displays the detected number along with the image

Chapter 3

System Analysis

3.1 Functional Requirements

The system performs the following functions based on the implemented ANPR pipeline:

- **Input Image Handling**

  The system accepts a vehicle image from the local images folder.

- **Image Preprocessing**

  The input image is converted into grayscale and filtered using a bilateral filter to reduce noise while preserving edges.

- **Edge Detection**

  Canny Edge Detection is applied to identify edges and highlight the boundaries of objects, especially the number plate.

- **Contour Detection**

  The system detects contours from the edge-detected image and selects the most suitable contour representing the number plate based on shape

- **Number Plate Localization**

  The detected contour is used to locate the exact position of the number plate in the image.

- **Masking and Cropping**

The system creates a mask to isolate the number plate region and extracts it for further processing.

- **Text Recognition using EasyOCR**

The cropped plate image is passed to EasyOCR, which extracts the alphanumeric characters from the plate.

- **Result Visualization**

The system draws a bounding box around the detected plate and displays the recognized text on the image.

- **Console Output**

The detected number plate text is printed in the terminal.

3.1.2 Non-Functional Requirements

- **Performance**
  The system processes a single image within a few seconds.
- **Accuracy**
  Provides moderate accuracy for clear images; accuracy may decrease with blur, angle, or poor lighting.
- **Usability**
  The system is simple to use and requires minimal user interaction.
- **Reliability**
  The system performs consistently for standard number plate images.
- **Scalability**
  The system can be extended to real-time video processing or integrated with advanced models.

3.2 Hardware Requirements

- **Processor**: Intel i3 or higher
- **RAM**: Minimum 4GB (8GB recommended)
- **Storage**: 10–20GB free space
- **GPU**: Not required

3.3 Software Requirements

- **Operating System**: Windows
- **Programming Language**: Python 3.x
- **Libraries Used**:
    - OpenCV (Image processing)
    - EasyOCR (Text recognition)
    - NumPy (Array operations)
    - Matplotlib (Visualization)
    - Imutils (Contour processing)
- **Development Tool**:
    - Visual Studio Code
- **Environment**:
    - Python Virtual Environment (ml-env)
- **Database**:

  Not used in the current implementation

Chapter 4
System Design

4.1 System Architecture

The proposed system follows a **pipeline-based architecture** instead of a web-based architecture. It processes an input image step-by-step to detect and recognize the number plate.

Architecture Flow:

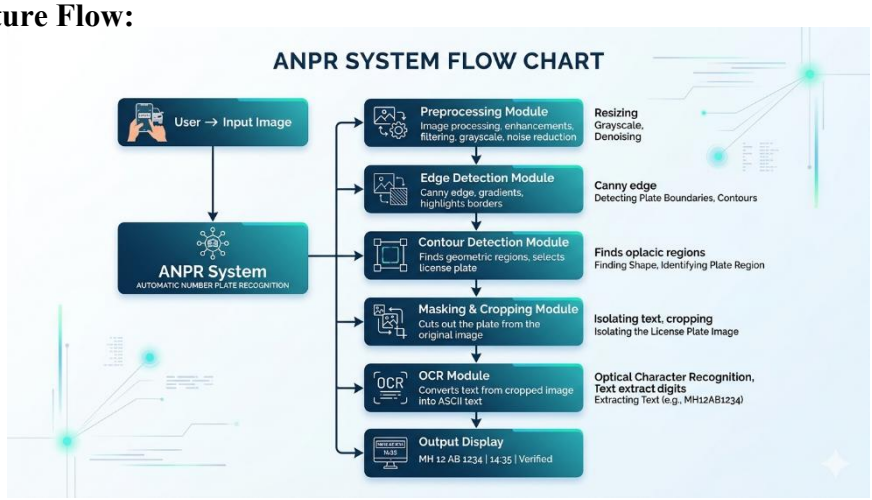

Figure 1 Architectural Flow of the ANPR System

Description:

- The user provides an input image.
- The system processes the image through multiple modules.
- Each module performs a specific task.
- Final output is displayed with detected number plate text.

4.2 Data Flow Diagram (DFD)

Level-0 DFD

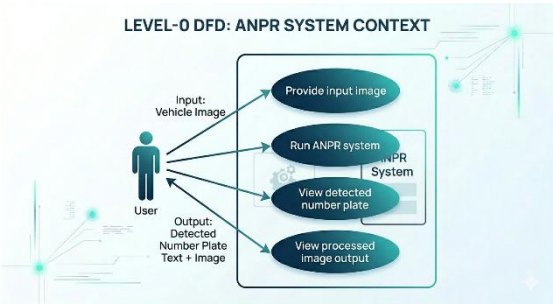

Figure 2 Level-0 DFD of the ANPR System

Explanation:

- The user provides an image.
- The system processes it internally.
- The output is the detected number plate and processed image.

Level-1 DFD

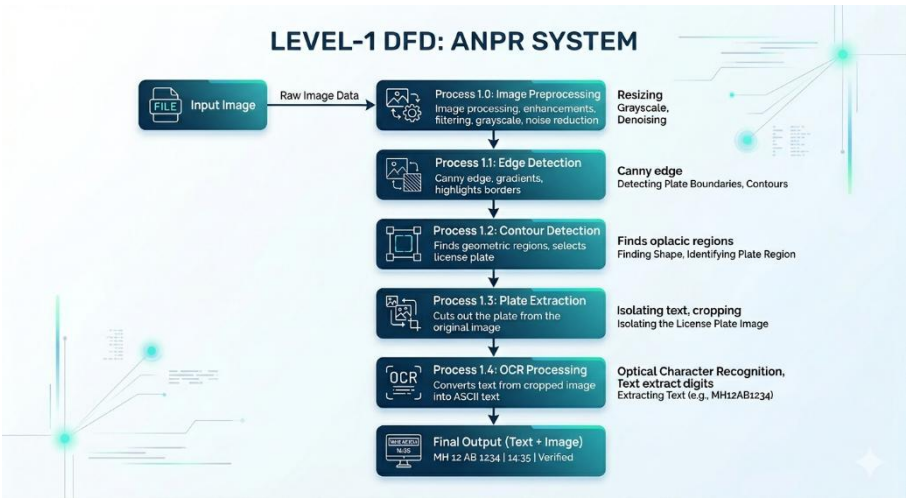

Figure 3 Level-1 DFD of the ANPR System

Explanation:

- The system is divided into modules.
- Each module transforms the data step-by-step.
- Final output is generated after OCR processing.

4.3 Use Case Diagram

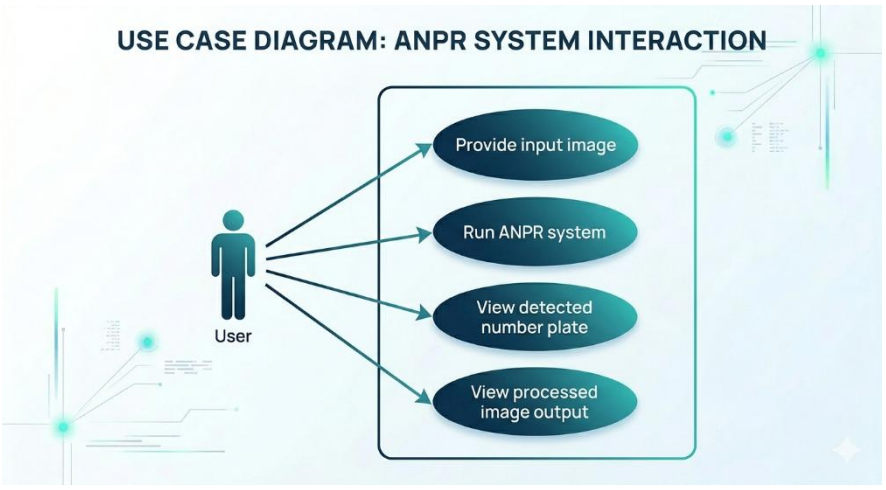

Figure 4 Use case Diagram of the system

Use Cases:

- Provide input image
- Run ANPR system
- View detected number plate
- View processed image output

Description:

- The user interacts with the system by providing an image.
- The system processes the image automatically.
- The user receives the detected number plate as output.

4.4 Database Design

Database Usage:

- No database is used in the current implementation.

Reason:

- The system is designed as a standalone image-processing application.
- It processes images and directly displays results without storing data.

.

Chapter 5

System Implementation

5.1 Frontend Implementation

In this project, no traditional frontend (such as HTML, CSS, or React) is used. The system is implemented as a standalone Python application.

- The user interacts with the system through the **Python script execution**.
- Images are provided through a local folder (images).
- Output is displayed using **Matplotlib**, which shows:
  - Original image
  - Processed image
  - Final result with bounding box and detected text

5.2 Backend Implementation

The backend of the system is fully implemented in Python using multiple modules.

The system is divided into the following components:

- **Preprocessing Module**

  Converts the image into grayscale and applies bilateral filtering to remove noise.

- **Edge Detection Module**

  Uses Canny Edge Detection to identify edges in the image.

- **Contour Detection Module**

Detects contours and selects the most appropriate one representing the number plate.

- **Masking and Cropping Module**

  Extracts the number plate region from the image using masking techniques.

- **OCR Module**

  Uses EasyOCR to recognize and extract text from the cropped number plate.

- **Visualization Module**

  Draws a bounding box around the detected plate and displays the recognized text.

5.3 Database Implementation

No database is used in the current implementation.

- The system processes images in real time and directly displays results.

  The results are stored in excel locally

Plate Number	Timestamp
KA03AB3289	53:59.1
6H6517A	06:06.6
51RL4525	09:51.4
1SUYAB1234	12:52.1
KLO7AH9981	13:53.9
ABC1234	22:07.5

Table 1 Detected plates and timestamp

Future Enhancement:

A database can be added to:

- Store detected vehicle numbers
- Maintain logs of processed images

- Enable search and tracking features

5.4 Key Features Implementation

The following key features are implemented in the system:

- **Image Input System**

  Allows the user to provide vehicle images for processing.
- **Automatic Number Plate Detection**

  Uses contour-based techniques to locate the number plate.
- **Image Enhancement**

  Applies filtering and thresholding to improve OCR accuracy.
- **Text Recognition using OCR**

  Extracts alphanumeric characters from the number plate using EasyOCR.
- **Bounding Box Visualization**

  Highlights the detected number plate with a rectangle.
- **Text Display**

  Displays the detected number both on the image and in the console.
- **Modular Design**

  Each functionality is implemented in separate Python files for better organization.

Chapter 6

System Testing

6.1 Introduction of Testing

System testing is performed to ensure that the ANPR system works correctly and produces accurate results under different conditions. The testing focuses on verifying each module such as image preprocessing, edge detection, contour detection, and OCR.

Different test cases were executed using various input images to evaluate the system's performance and accuracy.

6.2 Test Cases

Test Case	Input	Expected Output
Image Loading	Valid image file	Image loaded successfully
ImageLoading (Invalid)	Wrong image path	Error message displayed
Preprocessing	Input image	Grayscale and filtered image displayed
Edge Detection	Processed image	Edges highlighted correctly
Contour Detection	Edge-detected image	Plate contour detected
ContourDetection Failure	Image without clear plate	No contour detected
Masking and Cropping	Detected contour	Plate region extracted correctly
OCR Detection	Clear number plate image	Correct text extracted
OCR Partial Detection	Blurred/low-quality image	Partial or incorrect text
Final Output Display	Valid processed image	Bounding box and detected text displayed

Table 2 Test cases of the ANPR System

6.3 Performance Testing

System Response Time

- The system processes a single image within **2 to 5 seconds**, depending on image size and system configuration.
- The first execution may take longer due to the initialization of the EasyOCR model.
- Subsequent executions are faster as the model is already loaded.
- Image preprocessing and contour detection steps execute quickly, while OCR takes comparatively more time.

Load Handling

- The system is designed to process **one image at a time**.
- It performs efficiently for small-scale usage with individual image inputs.
- The current implementation does not support batch processing or handling multiple images simultaneously.
- For large-scale or real-time applications, further optimization or parallel processing techniques would be required.

Chapter 7
Results and Discussion

7.1 Results

The developed Automatic Number Plate Recognition (ANPR) system was tested on various vehicle images to evaluate its performance. The system successfully detected and recognized number plates in most of the test cases.

Output Screenshots

The following outputs were generated during execution:

- Original input image
- Pre-processed image (grayscale and filtered)
- Edge-detected image
- Masked and cropped number plate
- Final output image with bounding box and detected text

Figure 5 Detecting the Number plate

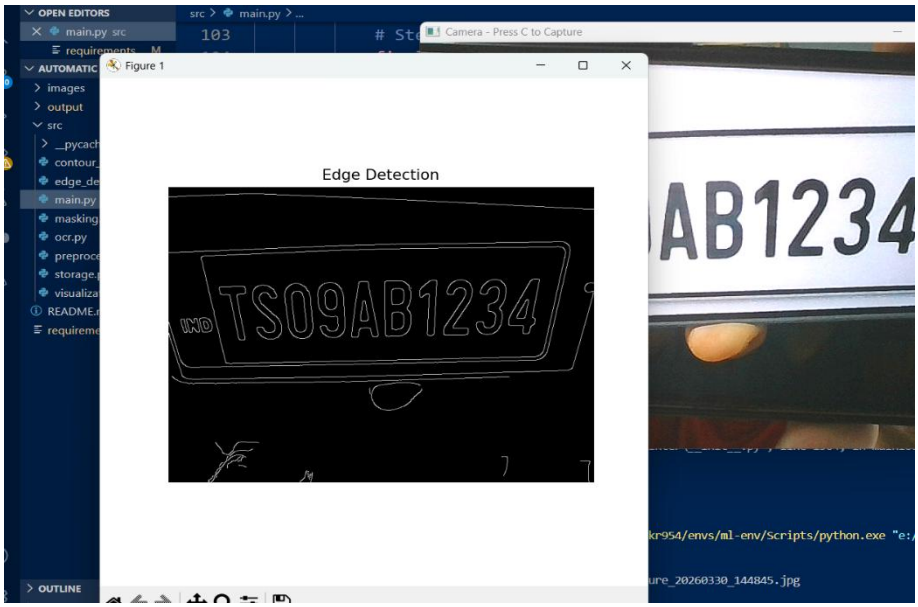

Figure 6 System highlights the number plate region with a rectangle

The system highlights the number plate region with a rectangle and displays the recognized text on the image as well as in the console.

Sample Output

- Detected Number Plate Text: Example – *HR26BR9044*
- Bounding box correctly drawn around the number plate
- Cropped plate image clearly visible after masking

Figure 7 Actual plate is detected and cropped

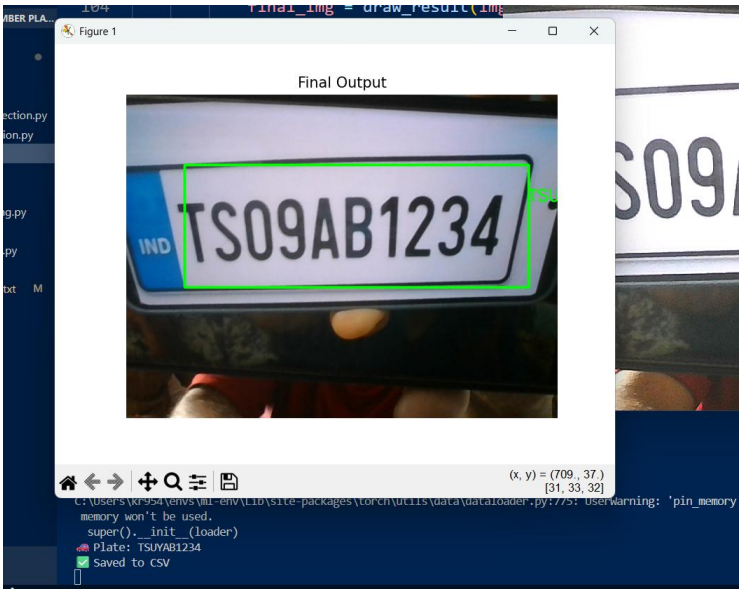

Figure 8 Actual plate is detected and number is stored in Excel file

7.2 Discussion

The performance of the system depends on several factors:

Accuracy Analysis

- The system provides **good accuracy for clear and high-resolution images**.
- It can successfully detect rectangular number plates using contour detection.

- EasyOCR is capable of extracting text without prior training.

Limitations Observed

- OCR may detect partial text (e.g., “IND” only) in some cases.
- Performance decreases with:
    - Blurred images
    - Poor lighting conditions
    - Angled or rotated number plates
- Contour-based detection may fail if the plate is not clearly distinguishable.

System Behavior

- Preprocessing improves image quality and helps in better detection.
- Edge detection plays a key role in identifying plate boundaries.
- OCR accuracy improves when the cropped image is enhanced (resizing and thresholding).

Overall Performance

- The system works efficiently for basic ANPR tasks.
- Suitable for small-scale applications and learning purposes.
- Provides a foundation for further improvements using advanced techniques.

7.3 Summary

- The system successfully demonstrates number plate detection and recognition.
- Results are satisfactory for controlled conditions.
- Further improvements can enhance accuracy and robustness.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

The developed Automatic Number Plate Recognition (ANPR) system successfully demonstrates the use of computer vision and Optical Character Recognition (OCR) techniques to detect and recognize vehicle number plates from images. The system automates the process of identifying number plates, reducing the need for manual effort and minimizing human errors.

By using image preprocessing, edge detection, contour detection, and EasyOCR, the system is able to locate the number plate region and extract the corresponding text effectively. The modular design of the system makes it easy to understand, implement, and extend for future enhancements.

The project achieves its main objective of building a functional ANPR system that works on static images and provides satisfactory results under normal conditions. Although the accuracy may vary depending on image quality, lighting, and angle, the system serves as a strong foundation for real-world applications.

Overall, the project highlights the practical implementation of computer vision techniques and provides a base for further development into more advanced and real-time number plate recognition systems.

8.2 Future Work

The current ANPR system provides a basic and functional solution for detecting and recognizing number plates from static images. However, there are several areas where the system can be improved and extended for real-world applications.

- **Real-Time Video Processing**
  The system can be enhanced to process live video streams from cameras for real-time number plate detection and recognition.

- **Integration with Deep Learning Models**
  Advanced models such as YOLO can be used for more accurate and robust number plate detection, especially in complex environments.
- **Improved OCR Accuracy**
  Custom-trained OCR models or deep learning-based text recognition can be implemented to improve accuracy for different fonts and plate formats.
- **Database Integration**
  A database can be added to store detected number plates, timestamps, and logs for tracking and analysis.
- **Web or Mobile Application**
  The system can be converted into a web or mobile application for easier accessibility and user interaction.
- **Automatic Traffic Monitoring System**
  The system can be integrated with traffic management systems to automatically detect violations such as over-speeding or unauthorized vehicles.
- **Multi-Image and Batch Processing**
  Support for processing multiple images simultaneously can be added for large-scale applications.
- **Security and Surveillance Integration**
  The system can be used in parking areas, toll booths, and restricted zones for automated vehicle monitoring.

References

1. IEEE Style (most commonly used in engineering papers)
A. S. Agbemenu, J. Yankey, and E. O. Addo, "An automatic number plate recognition system using OpenCV and Tesseract OCR engine," *International Journal of Computer Applications*, vol. 180, no. 43, pp. 1–5, May 2018.

2. APA Style (7th edition)
Agbemenu, A. S., Yankey, J., & Addo, E. O. (2018). An automatic number plate recognition system using OpenCV and Tesseract OCR engine. *International Journal of Computer Applications, 180*(43), 1–5.

Appendix

API documentation

The ANPR system uses several Python libraries and APIs for image processing and text recognition.

1. OpenCV (cv2)

OpenCV is used for image processing and computer vision operations.

Functions Used:

- cv2.imread() – To read the input image
- cv2.cvtColor() – To convert image from BGR to grayscale/RGB
- cv2.bilateralFilter() – To reduce noise while preserving edges
- cv2.Canny() – To perform edge detection
- cv2.findContours() – To detect contours in the image
- cv2.drawContours() – To draw contours on the image
- cv2.rectangle() – To draw bounding box around number plate
- cv2.putText() – To display detected text on image

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2. EasyOCR

EasyOCR is used for Optical Character Recognition to extract text from the number plate.

Functions Used:

- easyocr.Reader() – To initialize OCR model
- readtext() – To extract text from image

---

3. NumPy

NumPy is used for numerical operations and image array handling.

Functions Used:

- np.zeros() – To create mask image
- np.where() – To locate plate region coordinates
- np.min() / np.max() – To find bounding box limits

4. Matplotlib

Matplotlib is used for displaying images.

Functions Used:

- plt.imshow() – To display images
- plt.title() – To add title
- plt.show() – To show output window
-

5. Imutils

Imutils is used to simplify contour handling.

Functions Used:

- imutils.grab_contours() – To extract contours correctly